Project Proposal

for

InternX

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1. Problem Statement:

- Many students struggle to find internships while enrolled in degree programs.
- Most internships require prior experience, creating a barrier for beginners.
- Available opportunities are limited and highly competitive.
- Companies lack access to reliable and affordable entry-level talent for short tasks.
- There is no dedicated and structured platform that verifies student status, evaluates their skills, and efficiently connects them with suitable companies.

This project aims to address this problem by providing a secure, student-only internship platform that connects verified students with businesses for short-term, experience-focused tasks. The system promotes fairness through skill evaluation, price control, and progress tracking. Thus, students can gain valuable experience while businesses find affordable talent.

2. Project Scope:

This project focuses on developing a student-only internship and freelance platform designed to connect verified university students with businesses seeking affordable and reliable talent. The platform ensures that only currently enrolled BS/MS students can participate through strict verification using university email addresses or official student ID proof, with automatic account suspension after a defined period post-graduation.

2.1 Student & Business Interaction: Students can create detailed profiles, add limited and relevant skills, define availability, and work on only one project at a time to ensure fair opportunity. Businesses post projects with fixed difficulty levels and pricing rules, and can select from a limited number of eligible students, preventing market saturation and unhealthy competition.

2.2 Project Execution & Control: The platform supports hourly and fixed-price contracts, progress tracking through a dedicated project dashboard, paid communication meetings, and structured deliverable submission.

2.3 Reviews & Growth: Upon project completion, payments are released and mutual reviews are recorded. Student levels are updated automatically based on completed projects, supporting structured growth and skill progression.

3. Tools and Technologies:

Frontend

- **React.js** : For building a responsive and interactive UI.
- **HTML5, CSS3, JavaScript** : Core technologies for structure, styling, and interactivity.

Backend

- **Node.js with Express.js** : Handles REST APIs for authentication, projects, applications, and messaging.
- **JWT (JSON Web Tokens)** : Provides secure authentication and role-based access.

Database

- **MongoDB** : NoSQL database to store users, projects, skills, and progress tracking.

Artificial Intelligence & Machine Learning

- **Python** : Open-source language for AI modules. Runs locally or via **Google Colab**.
- **NLP & AI Test Generation** : Generates skill tests from project task descriptions.
- **AI-Based Skill Matching** : Recommends suitable internships to students based on verified skills, academic level, and availability.
- **Computer Vision** : Monitors skill tests using Python libraries (OpenCV, Mediapipe) to detect cheating.

Development Tools

- **Visual Studio Code** : IDE for coding.
- **Git & GitHub** : Version control and collaboration.
- **Postman** : API testing, for development use.

4. Review of Similar Existing Projects:

4.1 Internshala

India's largest internship platform with verified listings and training courses. Weaknesses: No AI skill verification, high competition, accepts graduates, focuses on long-term internships.

4.2 LinkedIn

Global professional network with internship features and AI recommendations. Weaknesses: Not student-specific, no skill testing or progress tracking, heavy competition.

4.3 Handshake

University-partnered platform with personalized recommendations.
Weaknesses: Limited to partner universities, no AI testing or anti-cheating measures.

4.4 Fiverr / Upwork

Freelancing platforms with task-based work and rating systems.
Weaknesses: Dominated by professionals, no student verification or skill testing, no educational focus.

4.5 Academic Research

Studies (Zhang et al., 2021; Kumar & Singh, 2022; Lee et al., 2023) explored AI matching and proctored testing with promising results.
Limitation: Remain theoretical without real-world platforms integrating verification, testing, and micro-internship tracking.

5. What Makes Our Project Different:

- **AI-Based Skill Testing Before Application:** Students must pass an AI-generated skill test before applying. The test is created directly from the internship task description, ensuring students cannot misrepresent their skills.

- **Live AI Monitoring to Prevent Cheating:** Skill tests are monitored through live camera-based AI analysis to detect cheating or external assistance, providing reliable and trustworthy skill verification for businesses.

- **Strict Student-Only Verification Model:** Access is limited to currently enrolled university students verified through official university emails or student ID checks. Accounts are automatically suspended six months after graduation, preventing professional freelancing dominance.

- **Micro-Internship-Centered Platform:** The system is designed exclusively for short, task-based internships (1–2 weeks), enabling fast skill acquisition for students and low-risk talent evaluation for companies.

- **Automated AI Skill Matching:** AI matches students to internships using verified skills, academic level, availability, and project difficulty, filtering out ineligible applicants and improving match accuracy.

- **Performance-Backed Skill Badges:** Skills are validated through monitored tests and completed tasks, allowing the platform to award credible skill badges based on actual performance rather than self-declared resumes.

REFERENCE LIST

- Zhang, L., Wang, Y., & Chen, H. (2021). Intelligent recommendation system for internship placement using collaborative filtering. *Journal of Educational Technology & Society*, 24(3), 156-169.
- Kumar, A., & Singh, R. (2022). Machine learning-based internship matching using NLP. *International Journal of Information Management*, 62, 102-115.
- Lee, S., et al. (2023). Computer vision-based proctoring for online assessments. *Computers & Education*, 185, 104-118.